

Project Initiation Document

Project name: Sustainable Travel Choices

Project Sponsor: Steve Battle

Project Team: Group 03

Document Control

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V.3.0	19/12/2025	Business Case	Mihiranga/ Padam
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Document Particulars

Review & Sign-Off

Authors (Responsible)

Name	Contribution
Mihiranga	Lead Author: Created the overall project plan, risk register, and schedule.
James	Co-Author: Assisted with Gantt Chart and finalized Appendices
Thinod	Co-Author: Defined the project background, technical scope, and boundaries
Rison	Co-Author: Established the initial document structure and project initiation framework
Padam	Co-Author: Developed the business case and justification for the chosen solution
Samir	Co-Author: Conducted the risk assessment and integrated the survey data analysis

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Support (Supportive)

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Name	Level of Review/Criteria
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Mihiranga	Technical Review: Verified the feasibility of the proposed Gantt Chart timeline

Distribution (Informed)

Name	Role
Steve Battle	Module Leader

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1 Project Definition

1.1 Purpose

The main purpose of this project is to design and build a **"Proof of Concept" (PoC)** for a new mobile solution called the **UWE Smart Travel Hub**.

Our research has confirmed that students struggle with two major issues. They are **high travel costs** and the hassle of checking multiple apps for different transport modes. This project aims to solve these problems by bringing everything buses, e-scooters, and carpooling—into one simple interface. Additionally, we want to test if a **"Green Points"** reward system can successfully motivate students and staff to switch to more sustainable travel habits.

1.2 Aims & Objectives

Project Goal - To design and develop a Coded Proof-of-Concept (PoC) for the "UWE Smart Travel Hub" that demonstrates how consolidating travel data and incentivising sustainable choices can effectively reduce student commuting costs and improve campus accessibility.

SMART Objectives:

1. To conduct and complete the **User Requirement Analysis** and Student Survey by **5th January 2026** to validate the critical barriers of high cost and information fragmentation.
2. To design high-fidelity **User Interface (UI) wireframes** for the "Smart Journey Planner" and "Green Points Dashboard" using design tools (e.g., Figma) by **15th February 2026**.
3. To develop a functional **Proof-of-Concept (PoC) prototype** that utilizes **Simulated Data** (JSON) for weather and bus timings to demonstrate the application's "Smart Brain" recommendation logic by **10th March 2026**.
4. To implement and test the **"Green Points" algorithm** within the prototype to demonstrate how sustainable travel choices (Walking/Cycling) are converted into redeemable reward points by **19th March 2026**.

1.3 Success criteria

To consider this project a success, we must deliver the following:

- **Working 'Smart Logic' Demo:** The prototype must successfully demonstrate that the recommendation engine actually works. It needs to dynamically change its suggestions based on the simulated data. For example, in the "**Rain Scenario**," the app should detect the bad weather and automatically suggest the Bus instead of Cycling.
- **Proven 'Green Points' System:** We must demonstrate that the rewards logic is fully functional within the prototype. The system needs to show clearly how a student's sustainable trip is tracked and converted into actual points or discounts, meeting the client's sustainability goals.
- **Data-Backed Validation:** The project must be supported by real evidence. We need to complete the User Requirement Analysis using our **survey data** to prove that our core assumptions specifically about high commuting costs and the confusion of using multiple apps are accurate.
- **On Time Delivery:** We must submit the complete Project Portfolio (including the PID, Technical Report, and the Coded Prototype) by the **April 2026** deadline, ensuring all work follows the university's academic and ethical guidelines.

1.4 Assumptions & constraints

Assumptions

Assumption 1: Incentives will work: Our survey data highlighted a big gap in carpooling: **64%** of students want a ride, but only **8%** are willing to drive. We assume that the "Green Points" reward system will be a strong enough motivator to encourage more car owners to offer rides, helping to balance this out.

Assumption 2: Independent User Authentication and Verification The project assumes that user identity and security can be effectively managed through an internal authentication gateway that validates the '@uwe.ac.uk' institutional domain during registration. By implementing this domain-restricted verification process, the system ensures that access is exclusively reserved for the UWE community. This standalone validation mechanism provides a secure environment for features like carpooling while eliminating the technical dependency on UWE IT Services or external Student ID database integration.

Assumption 3: Cost is the main driver We assume that the main reason students are unhappy with their current travel is the expense. Our survey confirmed that **60%** spend more than **£4.00 daily**. Therefore, we assume that if our app helps them save money, they will actively switch to using it.

Constraints

- **Technical Constraint (Live API Integration):** While the project operates under a zero-budget model, it will prioritize the integration of freely available third-party APIs, such as the **Bus Open Data Service (BODS)** and **Dott API**, to provide real-time travel information. To mitigate the risk of technical downtime or data unavailability during the final presentation, a pre-recorded system demo will be maintained as a backup. Paid APIs (e.g., Google Maps Premium) remain out of scope due to cost.
- **Scope Constraint (Prototype Only):** The final deliverable is strictly a **non-functional Proof-of-Concept (PoC)** designed to demonstrate the User Interface and logical algorithms. It is **not** a production-ready application capable of handling real user traffic or secure payments.

- **Time Constraint:** All project deliverables, including the Final Report and the Coded Prototype, must be completed and submitted by the non-negotiable academic deadline in **April 2026**.

1.5 Approach and Methodology

The project will follow a **Tailored Agile Approach**, incorporating specific elements from the *Agile Manifesto* (Beck et al., 2001) to suit the constraints of a 12-week academic project. This iterative methodology allows the team to build the Proof of Concept (PoC) in manageable steps, ensuring continuous testing of the "Smart Brain" logic and UI design based on ongoing feedback.

Key Agile Elements Utilized:

- **Sprints:** The timeline is divided into development iterations to manage tasks effectively.
- **Weekly Stand-up Meetings:** The team holds weekly reviews to track progress and resolve technical blockers.
- **Version Control:** Code is managed via a shared **GitHub** repository to ensure transparency and version history.

Project Phases:

- **Phase 1: Research, Analysis & Design (Weeks 1-6)**
 - Conducting User Requirement Analysis via the student survey (Qualtrics).
 - Developing High-Fidelity Wireframes (using Figma) to visualize the "Journey Planner" and "Green Points" dashboard.
- **Phase 2: Prototype Development (Weeks 7-12)**
 - **Frontend Development:** Coding the interface using HTML, CSS, and JavaScript.
 - **Hybrid Data Integration:** Integrating live third-party APIs (e.g., **BODS**, **Dott**) for real-time travel data, with static JSON files as a fallback.
 - **Algorithm Implementation:** Coding the "Green Points" logic for sustainable choices.
- **Phase 3: Testing, Evaluation & Closure (Weeks 13-16)**
 - **Scenario Testing:** Validating logic against use cases like the 'Rain Scenario'.
 - **Final Reporting:** Documenting the technical implementation and reflecting on the project outcome.

Reference: Beck, K. et al. (2001) *Manifesto for Agile Software Development*. [Online] Available from: <https://agilemanifesto.org/> [Accessed 4 March 2026].

2 Business Case

2.1 Background

The Current Situation Right now, planning a trip to UWE is more complicated than it should be. There is no single central app for students, so we are forced to jump between different services. Usually, a student has to check **First Bus** for times, **Google Maps** for directions, and **Dott** for scooters separately. Because there is no platform to bring all this together, it is hard to find the best option, and there is no system in place to reward students for choosing sustainable travel.

What the Data Says (Survey Analysis) We surveyed **UWE students** in January 2026 to understand the real issues. The results confirmed our suspicions:

- **Cost is the biggest issue:** Commuting is expensive. Our data shows that **53%** of students spend more than £4.00 per day, and **57%** told us that "Cost is high" is their single biggest frustration.
- **Too many apps:** The process is messy. **60%** of students said they have to use 2 or more apps just to plan one trip. As a result, students ranked "Real-time Bus & Traffic comparison" as the feature they want the most.
- **The Carpooling Problem:** There is a huge mismatch in ridesharing. While **57%** of students want to be passengers, only **13%** are willing to drive. This shows that without a strong incentive system, carpooling won't work.
- **The Solution:** There is a clear demand for change. **60%** of students said they would "Yes, definitely" switch to sustainable transport if a "Green Points" reward system was introduced.

2.2 Options Considered

Option 1: Do Nothing (Keep things as they are)

- **Description:** Continue with the current situation where students have to juggle multiple apps (First Bus, Maps, Weather) to plan a trip.
- **Decision: REJECTED.**
- **Reasoning:** We rejected this because it ignores the problems we found. Our survey clearly showed that **60%** of students are struggling with high travel costs. Doing nothing would mean leaving them with this financial burden. Also, sticking to the old way doesn't help the University achieve its sustainability goals.

Option 2: Build a Full App with Live Data

- **Description:** Attempt to build a production-ready app that connects to real-time Google Maps and First Bus APIs.

- **Decision: REJECTED.**
- **Reasoning:** This is too risky for a 12-week student project.
 - **Cost:** Commercial APIs (like Google Maps) cost money, and we have a zero budget.
 - **Complexity:** Setting up secure payments and live tracking is too complex to finish perfectly within the semester.
 - **Risk:** If an external server goes down during our final presentation, our demo would fail. We can't take that risk.

Option 3: Proof-of-Concept (PoC) with Hybrid Live Data Integration

Description: Develop a working prototype that integrates free, live third-party APIs (such as BODS for bus data and Dott for micro-mobility) to provide real-world travel information.

Decision: SELECTED.

Reasoning: This is the most impressive and realistic approach as it moves beyond static "fake" data by utilizing available free data feeds to enhance technical complexity. It demonstrates high credibility while remaining within the zero-budget constraint.

- **Risk Mitigation (Safe Demo):** To prevent failure due to potential connection issues during the live presentation, the system will include a fallback mechanism using local JSON data and a pre-recorded video demo.
- **Zero Cost:** Utilizes only public, no-cost APIs, adhering to the project's financial boundaries.
- **Technical Validation:** Showcases the team's ability to handle real-world API integration and "Smart Brain" logic simultaneously.

2.3 Benefits and Costs

This project is designed to deliver real, practical benefits for students and the university, based directly on the needs identified in our survey:

- **Helping Students Save Money:** Our primary goal is to lower commuting costs. The survey revealed that **60% of students** are currently spending over **£4.00 per day** on travel. By highlighting the cheapest routes and making carpooling easier, the app aims to help students reduce this daily expense.
- **Saving Time:** Currently, **64% of students** have to switch between multiple apps (Google Maps, First Bus, Weather, Dott, etc...) just to plan one trip. Our "Smart Hub" solves this frustration by bringing all that data into a single screen, allowing students to plan their journey in seconds rather than minutes.
- **Encouraging Greener Travel:** We found that **68% of students** are willing to switch to sustainable transport if they get rewards. The "Green Points" system is designed to capture this interest, motivating students to leave their cars at home in exchange for rewards, which directly supports UWE's sustainability goals.
- **Better Access to Campus:** There is a huge gap in carpooling: while **64%** of students want a ride, only **8%** are offering one. By using incentives to encourage more drivers, we hope to unlock more transport options for students living in areas with poor bus connections.

Project Costs

The project will operate under a strict **Zero-Financial Budget** model, utilizing academic licenses and open-source tools.

- **Monetary Costs (£0.00):**
 - **Software/Tools:** Utilization of free educational licenses for **Figma** (UI Design), **Qualtrics** (Survey), and **GitHub** (Version Control).
 - **Data Feeds:** Usage of **Simulated Data (JSON)** eliminates the need for expensive paid API subscriptions (e.g., Google Maps Platform, OpenWeatherMap).
 - **Hosting:** The prototype will be hosted locally or via free static hosting services (e.g., GitHub Pages).
- **Non-Monetary Costs (Human Resources):**
 - The primary cost is the **"Sweat Equity"** (Time and Effort) of the project team.
 - **Estimated Effort:** Approximately **12-15 hours per week** per team member over the 12-week duration, covering research, design, coding, and documentation.

2.4 Risk Analysis

Risk Description	Likelihood	Impact	Mitigation Strategy
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Supply-Demand Imbalance in Carpooling: Survey data (Jan 2026) indicates a critical shortage of drivers (only 13%) compared to passengers (64%). This creates a risk where the "Ride-Share" feature fails due to a lack of available rides.	High	High	Dynamic Incentives: The "Green Points" algorithm will be adjusted to offer Double Points for Drivers compared to Passengers. This higher reward tier is designed to motivate more car owners to participate.
Scope Creep: There is a risk of adding too many complex features (e.g., Live Payments, GPS Tracking) which could exceed the strict 12-week academic timeline.	Medium	High	Strict Scope Management: The team will adhere to the "MoSCoW" prioritization method. Any feature not listed as "Must Have" in this PID is strictly out of scope. We will focus solely on the PoC and not a production-ready app.
Technical Skill Gaps: Team members may lack advanced proficiency in JavaScript or Figma, potentially delaying the UI/UX design and Logic implementation phases.	Medium	Medium	Pair Programming & Simplicity: The team will adopt a "Pair Programming" approach for complex tasks. We will also use simplified "Simulated Data" (JSON) instead of complex backend databases to match the team's skill level.
Unrealistic Simulation: Since Live APIs are excluded, there is a risk that the "Simulated Data" (Mock JSON) becomes too static, failing to convincingly demonstrate the "Smart	Low	High	Scenario-Based Datasets: We will create multiple JSON data files representing specific scenarios (e.g., rainy_day.json, heavy_traffic.json) to dynamically switch contexts during the demo, proving the logic works under different conditions.

Brain" logic during the final presentation.			
Team Resource Availability: Risk of delays due to team members falling ill or having conflicting deadlines with other modules.	Medium	High	Agile-Lite Workflow: Weekly stand-up meetings will track progress. All code will be hosted on a shared repository (GitHub) so any member can pick up another's work if an absence occurs.

3 Project Scope

3.1 Scope – inclusions and exclusions

Inclusions (What IS included in the project)

We are focusing on delivering a working **Proof-of-Concept (PoC)** web prototype. The project scope covers these specific areas:

- **Frontend User Interface (The Visuals):**
 - We will start by designing the layouts in **Figma** and then build a responsive web interface using **HTML & CSS**.
 - This includes coding the actual screens for the "Smart Journey Planner," the "Green Points" rewards page, and the "Ride-Share" dashboard.
- **Core Functionality (The 'Smart Brain'):**
 - We will code the recommendation logic that compares Cost, Time, and Weather to dynamically suggest the best transport mode.
 - **Identity Verification:** To verify users without a real database, we will simulate security checks. The system will only allow emails ending in @uwe.ac.uk and will check a code flag (isVerified: true) in our mock data before showing Carpooling options.
- **Data Simulation (The Backend):**
 - **Live Data Integration:** The prototype will connect to free, public transport APIs to fetch real-time bus timings and micro-mobility availability.
 - **Data Fallback Mechanism:** Static JSON files will be developed solely as a secondary fallback to ensure the "Smart Brain" logic can still be demonstrated if live external services are unreachable.

Exclusions (What is NOT included in the project):

To ensure we can successfully deliver this project within the 12-week deadline and the zero-budget constraint, we have specifically excluded the following:

- **No Commercial/Paid APIs:** While we are integrating free, open-source Live APIs (such as OSRM for routing and Open-Meteo for weather), we will NOT use commercial services like Google Maps Platform or premium tier weather APIs due to budget constraints.
- **No Dedicated Backend Server (Raw XML/GTFS Processing):** We will not build a Node.js or Python backend server to directly parse the complex raw XML/GTFS data feeds from the

First Bus or BODS services. Instead, our frontend-only Proof of Concept utilizes a custom JavaScript "Smart Matcher" algorithm to simulate bus route allocations dynamically based on live GPS coordinates.

- **No Live Payments:** We are not building a real checkout system. There will be no integration with payment gateways like Stripe or PayPal. The app will display ticket prices for informational and comparison purposes only.
- **Not a Native Mobile App:** We are developing a Responsive Web Application accessible via browsers (Chrome/Safari). We are not writing native code (Swift for iOS or Kotlin for Android) as it falls outside the project timeline and current team skill set.
- **No Real Student Data (GDPR Compliance):** We will not collect, process, or store any real personal or location data of UWE students on an external database. To ensure strict adherence to GDPR and ethical guidelines, the prototype will solely utilize "Mock User Accounts" for demonstration purposes.

3.2 Deliverables

User Requirement Analysis Report (Based on Survey Data)	By Mihiranga / Thinod	Phase 1: Research
UI/UX Wireframes (Figma) (Visual designs for Journey Planner & Green Points)	By James / Mihiranga	Phase 1: Design
Functional Web Prototype (PoC) (HTML/JS with Simulated JSON Data)	By Mihiranga / James / Thinod / Rishon	Phase 2: Development

To ensure the project, the team will deliver:

- **Documentation:** A Final Technical Report that critically analyses the survey findings.

- **Technical Product: High-Fidelity Wireframes** (Figma) and a **Functional Web Prototype** powered by simulated JSON data (excluding live APIs).
- **Demonstration: A Final Presentation** showcasing the "Smart Brain" logic and "Green Points" algorithm in action to verify the success criteria.

3.3 Constraints

We are working within several strict boundaries that have shaped our decision to build a "Proof-of-Concept" rather than a fully finished product:

- **Time Constraint (The Hard Deadline):** We are working to a fixed university timeline. Every part of this project including the final Code and the Report must be submitted by **23rd April 2026**. This deadline is non-negotiable, so any feature that takes too long to build has been removed from the plan.
- **Budget Constraint (£0.00 Budget):** We have absolutely **no financial budget**. This limitation forces us to be creative:
 - We are using **Open-source software** (GitHub, VS Code) instead of paid tools.
 - We are using **Simulated Data (JSON)** because real APIs like Google Maps charge per click.
 - We will host the site on free platforms (like GitHub Pages) rather than paid cloud servers.
- **Technical Constraint (Our Skills):** We have chosen technologies that match our team's current skills. We are building a **Responsive Web App** using HTML, CSS, and JavaScript. We decided *not* to build a Native Mobile App (iOS/Android) because learning new languages like Swift or Kotlin would take more time than we have available.
- **Data Protection Constraint (Ethics):** Following our **Ethical Review Checklist**, we are strictly forbidden from collecting real personal data from students. To stay compliant with GDPR, the system will only use **"Live User Accounts"** (fake data) for all demonstrations.

- **Resource Availability (Time Management):** We are a student team managing this project alongside our other academic modules. We estimate we can dedicate about **15 hours per week** each. This means we have to keep the project scope simple and focused on the core features only.

3.4 Assumptions

This project based on a few key assumptions. If any of these turn out to be wrong, we might need to adjust our plan.

- **Assumption 1: The Rewards will actually work** Our survey showed that **60%** of students are interested in rewards. We are building this app on the assumption that this interest isn't just talk—that the "Green Points" system will *actually* motivate students to change their daily habits. We also assume that offering double points to drivers will be enough of a "nudge" to fix the shortage of willing drivers.
- **Assumption 2: Our sample size is accurate enough** We know that **34 students** is a small sample size compared to the whole university. However, we are assuming that their feedback is representative of the wider population and that "High Cost" and "Too Many Apps" are indeed the main problems for everyone.
- **Assumption 3: Continuous Availability of Free Data Services.** We assume that free public data services, such as BODS, will remain accessible throughout the development phase. Should these services experience downtime, we assume that a recorded video demonstration of the live functionality will be an acceptable alternative for evaluation
- **Assumption 4: Team Commitment** We assume that every team member can stick to the plan of contributing roughly **15 hours per week**. We are planning the schedule assuming there will be no major personal emergencies or disruptions during these 12 weeks.

4 . Communications Plan

This section outlines the internal and external communication framework designed to maintain transparency and ensure all stakeholders are aligned with the project's progress .

Channel	Intended Audience	Purpose	Frequency
Microsoft Teams	Project Team	File sharing, document collaboration (PID, Technical Report), and virtual meetings.	Daily / As needed
WhatsApp Group	Project Team	Quick, informal updates, deadline reminders, and availability checks.	Daily
Weekly Progress Report (UWE Email)	Module Leader (Client)	Formal reporting of project status using the Four-Box method to ensure consistent client oversight.	Weekly (Fridays)
In-person Lab Sessions	Client + Team	Hands-on demonstration of the prototype, verbal feedback, and blocker discussion.	Weekly (Scheduled)
Blackboard	Examiners	Final submission of the Project Portfolio, Source Code, and Documentation.	At Deadlines

Reporting Mechanism (The Four-Box Method)

To move beyond informal updates, the team will implement the **Four-Box Reporting Method** as discussed in class. This provides a high-level, scannable overview of the project's health each week.

Each weekly report will specifically address:

1. Progress (Accomplishments):

- Summary of tasks completed (e.g., successful integration of BODS API or finalized Figma wireframes).

2. Next Steps (Forward Planning):

- Immediate goals for the upcoming sprint (e.g., coding the 'Green Points' logic or setting up the API fallback mechanism).

3. Risks & Issues (Blockers):

- Identification of technical obstacles (e.g., API response delays) or resource constraints that may impact the timeline.

4. Decisions Needed (Consultation):

- Specific items requiring the Client's input (e.g., final approval of the domain-based registration logic).

Stakeholder Engagement

The project recognizes three primary stakeholder groups:

- **Primary Beneficiaries:** UWE Students who will use the app to reduce travel costs .
- **The Client/Supervisor:** Steve Battle, who ensures academic and ethical standards are met .
- **The Project Team:** Group 03, responsible for technical delivery and documentation.

Project team membership and roles

The following table outlines the core project team and their specific responsibilities for the delivery of the PoC.

Name	Role
Mihiranga Ponaduwa	Lead Developer
James Heung	UI/UX Designer
Rishon / Thinod	Business Analyst
Padam / Samir	Tester / Researcher

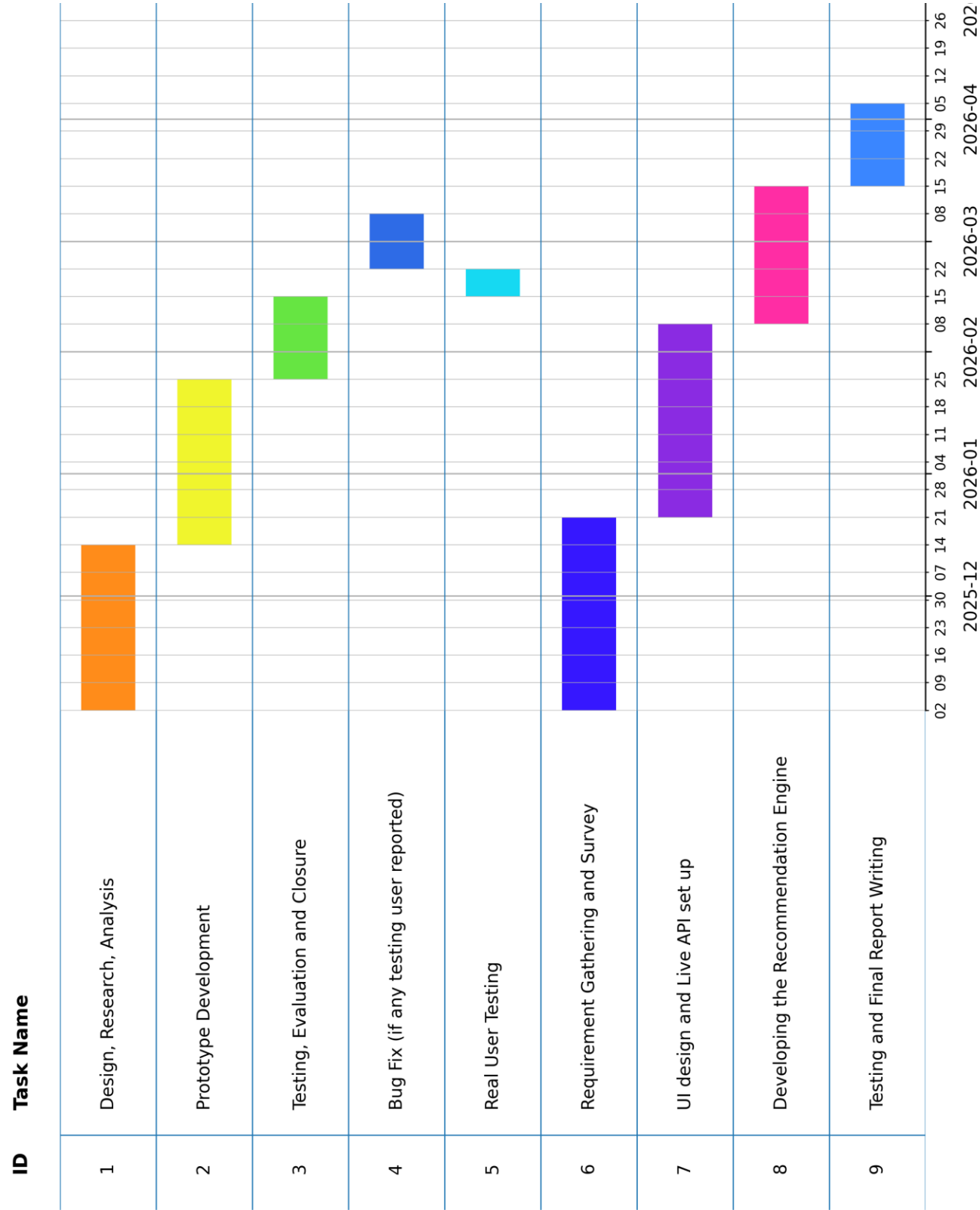
5 Appendices

Appendix A – related documents

The following project schedule outlines the critical path for the **UWE Smart Travel Hub** project. The timeline is structured into three main phases, ensuring all deliverables are met by the April 2026 deadline.

Phase	Task Description	Estimated Duration	Key Milestone
Phase 1	Research & Design (Requirement Gathering, Survey Analysis, UI Design)	Nov 2025 – Jan 2026	Survey Analysis Complete
Phase 2	Prototype Development (Coding Frontend, Setting up Live APIs, Recommendation Engine)	Jan 2026 – Mar 2026	Functional PoC Ready
Phase 3	Testing & Closure (User Testing, Bug Fixing, Final Report Writing)	Mar 2026 – Apr 2026	Final Project Submission

Design, Research, Analysis



Appendix B – glossary

Term / Acronym	Definition
API	Application Programming Interface. A set of functions that allows applications to access data and interact with external software components/operating systems.
Live API	An Application Programming Interface that fetches real-time, production-level data from external third-party sources (e.g., BODS for public transport timings and Dott for micro-mobility availability)
PoC	Proof of Concept. A realization of a certain method or idea in order to demonstrate its feasibility.
UI	User Interface. The point of human-computer interaction and communication in a device, including display screens and buttons.
UX	User Experience. The overall experience of a person using a product such as a website or computer application, especially in terms of how easy or pleasing it is to use.
Gamification	The application of game-design elements and game principles in non-game contexts (such as commuting) to improve user engagement.
Green Points	A proposed virtual reward currency within the app, awarded to users for choosing sustainable travel methods.
Micromobility	Transportation using lightweight vehicles such as e-scooters or bicycles, typically for short distances.

PID	Project Initiation Document. A logical document that brings together the key information needed to start the project on a sound basis.
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